GENE CONTROLLED TYPE ONE DIABETES, KNOW TREATMENT OF DIABETES EXCEPT INSULIN

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PROJECT SUMMARY

Latent autoimmune diabetes for adulthood study is to investigate the prevalence of autoimmune marker for certain diseases. The objective of MODY study is to identify the patients with MODY Genetic testing to confirm a clinical diagnosis of MODY has important implication for patients and their family. Diabetes Mellitus is a syndrome of chronic hyperglycemia, due to relative insulin deficiency, resistance or both. TYPE I diabetes belongs to family of HLA associated immune mediated organ specific disease.Auto antibodies directed against pancreatic islets constituents appear in the circulation within first few years of life. Auto antibodies are also found in older patients in LADA and carry an increased risk of progression to insulin therapy. More than 90% of the patient TYPE 1 diabetes carry HLA-DR3-DQ2, HLA, DR4, DQ8 or both as compared with 35% cases of background population. The greatest genetic contribution still comes from the HLA region modulated by a large number of genes with small effects. Insulin is coded by chromosome II and synthesized in β cells of pancreatic islets 50% secreted insulin is degraded in liver. The principle organ of glucose homeostasis is the liver. The brain is the major consumer glucose its function depends on uninterrupted supply of the substrate. Special glucose transporter GLUT protein carry glucose through membrane into cells. The insulin receptor is the gluco protein coded for short arm of chromosome 19. There are some rare form of diabetes. MODY (genetic defects of beta cell function) TNDM (developed in infants before 6 months occurs after birth and resolves within 12 weeks) PNDM (occurs soon after birth 50% of these case ultimately relapse later in life). There are some rare genetic cause of TYPE II 1) Maternally inherited diabetes and deafness 2) Wolfram syndrome

Diabetes is an extremely common disease. The mortality rate of diabetes is more than AIDS and breast cancer. Various other complication can arise from diabetes like obesity, hypertension etc. LADA is found about 10% on non insulin requiring patient and therefore far prevalent than classic TYPE I diabetes, so it is very important to gather knowledge about it. MODY requires correct and accurate methods of diagnosis. It helps to distinguish between TYPE I and TYPE II diabetes and avoid the unnecessary insulin in sulfonylureas treatment which severely affect patient health. There are different methods used for diabetes mellitus HbA/c test by fasting C peptide test, urine ketone body test, glutamic acid test, GAD antibody test, lipid profile test, creatinine test, sodium test, potassium test, oral glucose tolerance test. The monogenic and polygenic forms of diabetes are required for translational biology and integrative genemic research. Transcriptic and secretomic analysis related to MODY may provide a significant molecular insight in to disease and its procurement. The PSC is helpful to better understand the molecular mechanism underlying different form of MODY. LADA increases with increasing in age decade. These clinical phenotypic features of LADA are similar to type one diabetic patients and differs from DM2 patients.

KEYWORDS

LADA, MODY, HLA system, Insulin structure, Insulin secretion Glucose metabolism, Glucose utilization, Glucose transport, Hormonal regulation, Glucose production, Insulin receptor, Classification of MODY, Genetic origin of some rare form of diabetes, Importance of diabetes research, Gad antibdy test, Lipid profile test, Creatinine test, Potassium test, Sodium test, Urine for ketone body test, OGT test, Haemoglobin A1c test, Fasting C peptide test, Future perspective of diabetes.

OBJECTIVE OF THE PROPOSAL

Within the next decade, the genes that increase risk of developing of forms of diabetes will likely to be known. It is also important to know the advantage and disadvantages of predictive genetic diabetes. The ethical legal ans social issues associated with wide spread availability and use of predictive genetic test must been addressed. Ideally considerations of such issues will lead to the development of practice, guidelines for diabetes which will hopefully serve as a model for genetic testing for other complex diseases.

Here two case studies a) Latent Autoimmune diabetes for Adulthood (LADA) b) Maturation Onset diabetes for Young (MODY) is done.

The objective of Latent Autoimmune diabetes for Adulthood

The primary objective of this study is to investigate the prevalence of auto immune markers for pancreatitis, thyroiditis and CD in adults with Type 1 diabetes mellitus.

The objective of Maturation Onset diabetes for Young

The primary objective of this study is to identify patients with MODY and detect the MODY subtypes prevalence among the study subjects. Genetic testing to confirm a clinical diagnosis of MODY has important implications for both the patient and their family. Once the responsible mutation is detected the treatment and monitoring can be followed. Genetic counselling is also done.

INTRODUCTION

Diabetes Mellitus (DM) is a syndrome of chronic hyperglycemia due to relative insulin deficiency, resistance or both. It affects more than 220 million people world wide, and it is estimated that it will affect 440 million by the year 2030. Diabetes is usually irreversible and although patients can lead a reasonably normal lifestyle, its late complications result in reduced life expectancy and major health costs.

ORIGIN OF THE PROPOSAL

Type 1 diabetes belongs to a family of HLA associated immune mediated organ specific diseases. Genetic susceptibility is polygenic with the greatest contribution from the HLA region. Auto antibodies directed against pancreatic islets constituents appear in the circulation within the first few years of life. Auto antibodies are also found in older patients with LADA (Latent auto immune diabetes of adult) and carry an increased risk of progression to insulin therapy. Increased susceptibility to type 1 diabetes is inherited but the disease is not genetically predetermined. Identical twin of a patient with type 1 diabetes has a 30-50% chance of developing the disease which implies that non genetic factors must also be involved. The risk of developing diabetes by age 20 curiously is greater with a diabetic father (3-7%) than with a diabetic mother (2-3%). If one child in a family has type 1 diabetes, each sibling has a 6% risk of having diabetes.

HLA SYSTEM

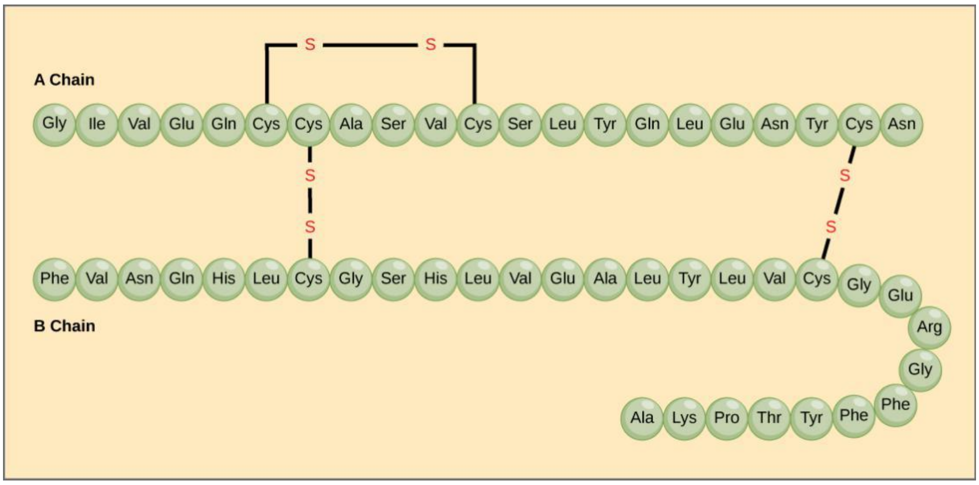
The HLA genes on chromosome 6 are highly polymorphic and modulate the immune defence system of the body. More than 90% of the patient with type 1 diabetes carry HLA-DR3-DQ2, HLA-DR4-DQ8 or both as compared with some 35% of the background population. Genotypic combinations have a major influence upon risk of disease. For example HLA-DR3-DQ2/HLA-DR4-DQ8 heterozygotes have a considerable inc reased risk of disease and some HLA class 1 alleles also modify the risk conferred by class 2 susceptibility genes.

OTHER GENES OR GENE REGION

Genome-wide association studies have greatly broadened our understanding of the genetic background to type 1 diabetes and more than 50 non-HLA genes or gene regions that influence risk have been identified to date. The greatest genetic contribution still comes from the HLA region, but this is modulated by a large number of genes with small effects. These include the gene encoding insulin (INS) on chromosome 11 and a number of genes involved in immune responses, including the cytotoxic T-lymphocyte-associated protein-4 (CTLA4) gene, the lymphoid-specific protein tyrosine phosphatage (PTPN22) gene and the IL-2R α-subunit of the IL-2 receptor complex locus (IL2RA), all of which are are implicated in a variety of HLA-associated autoimmune condition.

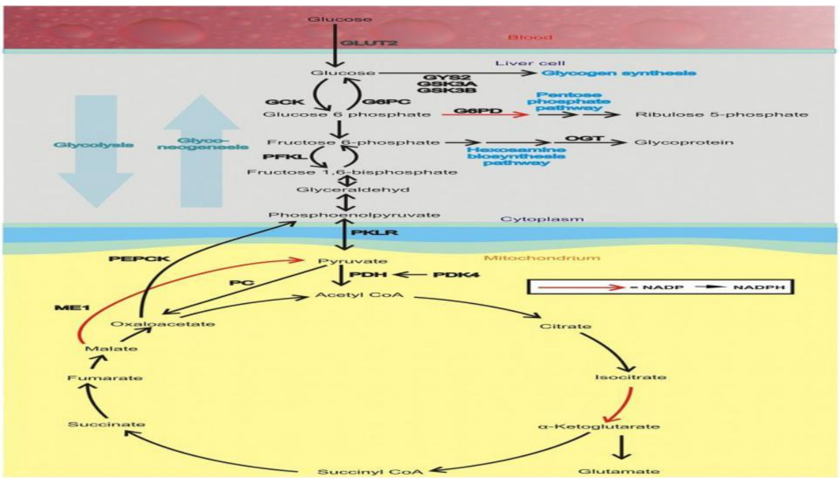
GENETICS AND DIABETES

INSULIN STRUCTURE AND SECRETION



Insulin is the key hormone involved in the storage and controlled release within the body of chemical energy available from food. It is coded for on chromosome 11 and synthesized in the beta cells of the pancreatic islets. The synthesis, intracellular processing and secretion of insulin by beta cells is typical of the way that the body produces and manipulates many peptide hormones. After secretion, insulin enters the portal circulation and is carried to the liver, its prime target organ. About 50% of secreted insulin is extracted and degraded in the liver, the residue is broken down by the kidneys. C-peptide is only partially extracted by the liver (and hence provides a useful index of the rate of insulin secretion) but mainly degraded by the kidneys.

GLUCOSE METABOLISM



Blood glucose levels are closely regulated in health and rarely stray outside the range of 3.5-8.0 mmol/L (63-144mg/dl), despite the varying demands of food, fasting and exercise. The principal organ of glucose homeostasis is the liver, which absorbs and stores glucose in the post absorptive state and releases it into the circulation between meals to match the rate of glucose utilization by peripheral tissues. The liver also combines 3-carbon molecules derived from breakdown fat (glycerol), muscle glycogen (lactate) and protein (like alanine) into the 6-carbon glucose molecule by the process of gluconeogenesis.

GLUCOSE PRODUCTION

About 200g of glucose is produced and utilized each day. More 90% is derived from liver glycogen and hepatic gluconeogenesis, and the remainder from renal gluconeogenesis.

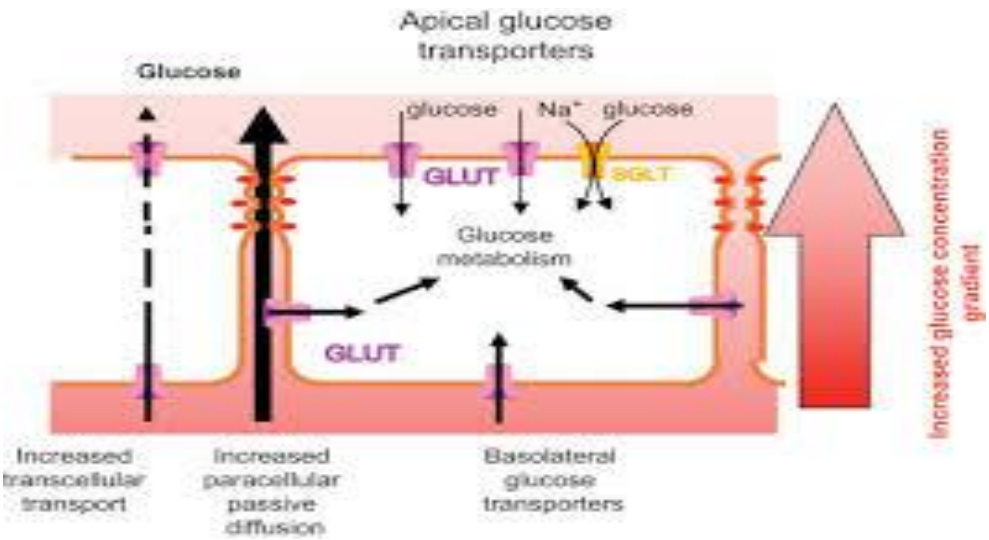
GLUCOSE UTILIZATION

The brain is the major consumer of glucose, and its function depends upon an uninterrupted supply of this substrate. Its requirement is 1mg/kg body weight per minute, or 100g daily in a 70kg man. Glucose uptake by the brain is obligatory and is not dependant on insulin, and the glucose used is oxidized to carbon dioxide and water. Other tissue, such as muscle and fat, are facultative glucose consumers. The effect of insulin peaks associated with meals is to lower the threshold for glucose entry into cells; at other times, energy requirements are largely met by fatty-acid oxidation. Glucose taken up by muscle is stored as glycogen or metabolized to lactate or carbon dioxide and water. Fat uses glucose as a source of energy and as a substrate for triglyceride synthesis; lipolysis releases fatty acids from triglyceride together with glycerol, a substrate for hepatic gluconeogenesis.

HORMONAL REGULATION

Insulin is a major regulator of intermediary metabolism, although its actions are modified in many respects by other hormones. Its actions in the fasting and postprandial states differ. In the fasting state, its main action is to regulate glucose release by the liver, and in the postprandial state, it additionally promotes glucose uptake by fat and muscle. The effect of counter-regulatory hormones (glucagon, epinephrine (adrenaline), cortisol and growth hormone) is to cause greater production of glucose from the liver and less utilization of glucose in fat and muscle for a given level of insulin.

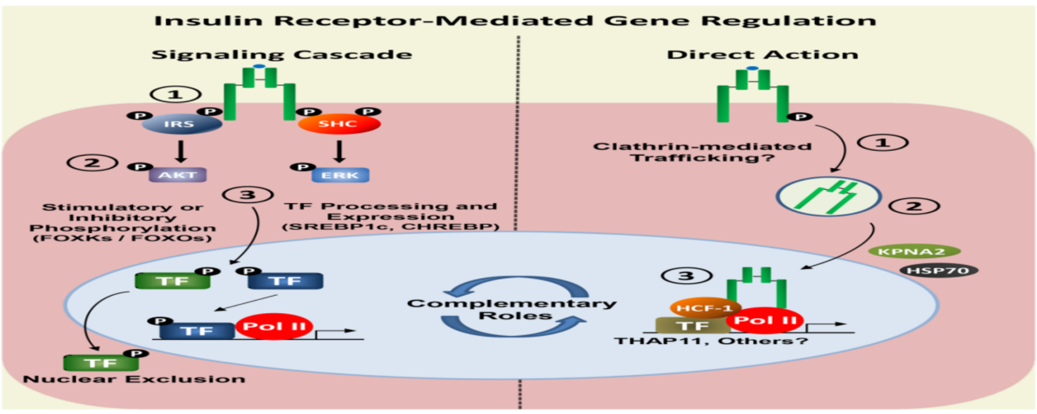
GLUCOSE TRANSPORT



Cell membranes are not inherently permeable to glucose. A specialized glucose-transporter (GLUT) proteins carry glucose through membrane into cells.

- GLUT-1:- enables basal non-insulin-stimulated glucose uptake into many cells.
- GLUT-2:- transports glucose into the beta cell, a prerequisite for glucose sensing, and is also present in the renal tubules and hepatocytes.
- GLUT-3:- enables non-insulin-mediated glucose uptake into brain neurones and placenta.
- GLUT-4:- enables much of the peripheral action of insulin. It is the channel through which glucose is taken up into muscle and adipose tissue cells following stimulation of insulin receptor.

THE INSULIN RECEPTOR



This is a glycoprotein (400kDa), coded for on short arm of chromosome 19, which straddles the cell membrane of many cells. It consists of a dimer with two α-subunits, which include the binding sites for insulin, and two β-subunits, which traverse the cell membrane. When insulin binds to the α-subunits it induces conformational change in the β-subunits, resulting in activation of tyrosine kinase and initiation of a cascade response involving a host of other intracellular substrates. One consequence of this is migration of the GLUT-4 glucose transporter to the cell surface and increased transport of glucose into cell. The insulin-receptor complex is then internalized by the cell, insulin is degraded, and the receptor is recycled to the cell surface.

GENETIC ORIGIN OF SOME RARE FORM OF DIABETES

Genetic defects of beta cell function (previously called "maturity onset diabetes of the young", MODY) are dominantly inherited, each associated with different clinical phenotypes. Infants who develop before 6 months of age are likely to have monogenic defect and not true type 1 diabetes ("transient neonatal diabetes mellitus", TNDM) occurs soon after birth and resolves at a median of 12 weeks. 50% of the case ultimately relaps.e later in life. This is called permanent neonatal diabetes mellitus (PNDM). TNDM have an abnormality of imprinting of the ZAC and HYMAI genes on chromosome 6q and PNDM is due to mutations in the KCNJ11 gene encoding the Kir6.2 subunit of the beta cell potassium ATP channel.

CLASSIFICATION OF MODY

MODY	GENE	CHROMOSOMAL LOCATION
MODY 1	HNF-4A	20q
MODY 2	Glucokinase	7p
MODY 3	HNF-1A	12q
MODY 4	Insulin promoter factor 1	13q
MODY 5	HNF-1B	17q

RARE GENETIC CAUSE OF TYPE 2 DIABETES

1. Maternally inherited diabetes and deafness : Mutation in mitochondrial DNA. Onset of diabetes before 40 years of age.

2. Wolfram's syndrome : Repressively inherited. Mutation in the trans membrane gene WFS1.

IMPORTANCE OF THE PROJECT

WHY IS DIABETES RESEARCH SO IMPORTANT?

Diabetes is an extremely common disease, affecting a diverse age range of people across the world. Those who are diagnosed with diabetes experience significant health concerns because the disease itself has proven to be the catalyst for other health problems. Many individuals who struggle with obesity develop diabetes. The diabetes also kills more people every year than breast cancer and AIDS combined.

Complications from diabetes can vary. However, the most prevalent co-morbid conditions include kidney disease, amputations, blindness, cardiovascular disease, obesity, hypertension, hypoglycemia, dislipidemia, and risk of heart attack or stroke.

Clinical trials allow researchers to carefully test diabetes medications before seeking FDA approval and releasing drugs to general public. Clinical trials offer researchers on important chance to compare and contrast blood glucose levels in patients, track metabolism, as well as monitor subsequent organ functionally related to medications. After that, researchers use this data to shape their unique drug formulas and to help increase the effectiveness of diabetes treatment. Some particular importance are studies to reduce environmental exposures during pregnancy or childhood that may increase diabetes and obesity risk. The incidence of type 1 diabetes is rising worldwide and the disease is occurring at younger ages suggesting an environmental trigger is responsible.

- ***IMPORTANCE OF LADA***

Despite the classification of diabetes into two major types, type 1 (insulin-dependent) diabetes and type 2 (non-insulin-dependent) diabetes, it is apparent that there are some forms of diabetes which do not fit comfortably into these categories. Indeed, there is one form of diabetes which appears to straddle the two major types, presenting with non-insulin requiring diabetes in adults, but with many of the genetic, immune and metabolic features of type 1 diabetes and with a high risk of progression to insulin dependency. This form of latent autoimmune diabetes of adults (LADA) is found in about 10% of initially non-insulin requiring diabetes patients and is therefore probably far more prevalent than classic type 1 diabetes. A major European Union initiative (ACTIONLADA) plans to learn more about it.

✧ *IMPORTANCE OF MODY*

Distinguishing MODY from other forms of diabetes represents an already available opportunity for so-called personalized or precision medicine as it creates an opportunity to select the treatment based on etiology. Insulin injections are the first line of type 1 diabetes treatment, while metformin is the first line of treatment for type 2 diabetes. On the other hand, *HNF1A*-MODY and sometimes *HNF4A*-MODY and are effectively treated with low-dose sulfonylureas, which are inexpensive oral diabetes medications.*GCK*-MODY has been shown to cause a mildly elevated baseline blood glucose that usually does not require pharmacologic management. The mild elevation of blood glucose does not lead to the common sequelae of diabetes such as nephropathy, neuropathy, microvascular or macrovascular complications.Personalized management of the 3 most common forms of MODY therefore results in improved patient care, through avoiding invasive insulin injections in favor of less expensive and more effective treatment methods. Multiple studies have demonstrated the improved patient experience resulting from a genetic diagnosis of MODY. A recent study modeling the cost-effectiveness of MODY genetic testing in all type 2 diabetes patients determined that the practice would surpass the cost-effectiveness thresholds that are used for resource-allocation decisions.In addition to the benefits of MODY genetic testing, genetic diagnoses for *KCNJ11* or *ABCC8* mutations in most NDM patients allows them to transition from insulin injections to high-dose sulfonylureas, which have dramatic effects on glycemic control. These clinical implications have provided the motivation for studies of MODY epidemiology in several different parts of the world, which have shown that MODY, while comparatively less common than T1DM and T2DM, affects an appreciable number of individuals (e.g., at least 1% of the nearly 30 million individuals with diabetes in the United States, or 300,000 people), and the vast majority of these individuals are not receiving a correct diagnosis.

METHODOLOGY

METHODS USED IN DIAGNOSIS DIABETES MELLITUS

Haemoglobin A1c (HbA1c) test for diabetes: The haemoglobin A1c test tells us our average level of blood sugar over the past 2 to 3 months. Its also called HbA1c, glycated hemoglobin test, and glyohemoglobin. People who have diabetes need this test regularly to see if their levels are staying within range. It can tell if we need to adjust your diabetes medicines. The A1c test is also used to diagnose diabetes. To measure a person's HbA1c level, a blood sample is taken from the patient's arm, and used to produce a reading. In some cases, such as with Hb1Ac testing for children, a single droplet of blood may be required to find out how much haemoglobin A1c is present. An HbA1c test may be used to check for diabetes or prediabetes in adults. Prediabetes means our blood sugar levels shows we are at risk for getting diabetes. If we already have diabetes, an HbA1c test can help monitor our condition and glucose levels.

Fasting C peptide test: The c-peptide test is a tool that the doctor uses to test whether we have type 1 diabetes, when the immune system attacks and destroys cells in the pancreas, or type 2 diabetes, when our body does not use insulin as well it should. It shows how well our blood makes insulin, which movies sugar from our blood into the cells.

A c-peptide test can be performed for any of the following reasons.

- To distinguish whether we may have type 1 or type 2 diabetes.
- To investigate whether we have insulin resistance.
- To ascertain causes of hypoglycemia (low blood glucose levels)
- To monitor insulin production after the removal of a tumor of the pancreas

To measure level of c-peptide a fasting blood test is taken.

We are asked not to eat or drink (certain fluids) for 8 to 12 hours before the test.

If we take blood glucose lowering medication we will likely be asked to stop taking these in the run up to the test.

For the blood test itself, a sample of blood will be taken from our arm and it shouldn't much more than a minute.

Level of c-peptide will be measured as well as the blood glucose level.

Urine for ketone body test: A ketone test can warn us of a serious diabetes complication called diabetic ketoacidosis, or DKA. An elevated level of this substance in our blood can mean we have very high blood sugar. Too many ketones can trigger DKA, which is a medical emergency. Regular tests we can take at home and can spot when our ketone bodies run too high. Then we can take insulin to lower our blood sugar level or get other treatment to prevent complications. A ketone test uses a sample of either our pee or blood.

	Blood ketone		
Urinary ketone	+	–	Total
+	14	5	19
–	29	74	103
Total	43	79	122
Sensitivity: 32.6%, specificity: 93.7%, PPV: 73.68%, NP: 71.84%			

❖ Urine test

We can buy this kind of test at our local drugstore and do it at home. Also, we can have one done while we are at doctor's chamber.

To take it, pee into a clean container to get a sample and then do the following:

- ➢ The strip is put from our test into the sample (or we can hold the test strip under our urine stream).
- ➢ The strip should be shaken gently.
- ➢ The strip will change colour; the directions will tell us how long that takes.
- ➢ The strip colour is checked against the chart that came with our test kit. This will show us the ketone level.

❖ Blood test

We can also take this test at home or at doctor's chamber. To take the blood sample, the doctor will put a thin needle into the vein in our arm to pull out blood or prick a finger. We can also use a home meter and blood test strips. Some blood glucose meters test for ketones too.

To take this kind of test at home:

- One of the blood ketone test strips is inserted into the meter until it stops.
- First hand is washed with soap and water, then it is dried.
- The finger is sticked with a lancing device.
- A drop of blood is placed into the hole on the strip.
- The result which is displayed on the meter is then checked.

Glutamic Acid Decarboxylase Autoantibodies test (GAD antibodies test): It is used to help discover whether someone has either type 1 diabetes or Latent Autoimmune Diabetes of Adulthood (LADA). A GAD antibody test may be favoured as a way of testing for which type of diabetes over a c-peptide test, which measures how much insulin is being produced by the body. It usually measures whether the body is producing a type of antibody which destroys its own GAD cells.

The GAD test is performed to help determine which type of diabetes someone has. The test is particularly useful for adults over 30 who get diabetes where a type 2 diabetes diagnosis is in doubt such as if the patient is not overweight. The test may be used to determine whether gestational diabetes (diabetes within pregnancy) may be type 1 diabetes. This test can also be used to measure the progression of type 1 diabetes or indicate a risk of type 1 diabetes or LADA.

To carry out GAD antibody test:

- A blood sample is taken from the patient's arm.
- The test should be done before insulin therapy is started.
- The blood sample will need to be sent to be analyzed by a lab before results are obtained.

Lipid profile test: The lipid profile test is a combination of tests conducted together to check for any risks in coronary heart disease, or as a preventive measure to check any risk depending on factors like eating habits, diet, stress, exercise and life style related. A typical profile includes the following tests

Lipid profile Test - Reference Values					
	Unit	Optimal	Intermediate	High
Total Cholesterol (calculated)	mg/dl	<200	200-239	>239
	mmol/L	<5.2	5.3-6.2	>6.2
LDL Cholesterol (calculated)	mg/dl	<130	130-159	>159
	mmol/L	<3.36	3.36 - 4.11	>4.11
HDL Cholesterol	mg/dl	>60	40 - 60	<40
	mmol/L	>1.55	1.03 - 1.55	<1.03
Triglycerides	mg/dl	<150	150 - 199	>1.99
	mmol/L	<1.69	1.69 - 2.25	>2.25
Non-HDL-C (calculated)	mg/dl	<130	130 - 159	>159
	mmol/L	<3.3	3.4 - 4.1	>4.1
TG to HDL Ratio (calculated)	mg/dl	<3	3.1 - 3.8	>3.8
	mmol/L	<1.33	1.34 - 1.68	>1.68

- High density lipoprotein cholesterol
- Low density lipoprotein cholesterol
- LDL/HDL ratio
- Triglycerides
- Very low density lipoprotein cholesterol

Test	Cara	Normal
Glucose, Serum (mg/dL)	82	65–99
Uric Acid, Serum (mg/dL)	4.2	2.5–7.1
Creatinine, Serum (mg/dL)	1.2	0.57–1.00
Sodium, Serum (mEq/L)	133	134–144
Potassium, Serum (mEq/L)	7.0	3.5–5.2
Chloride, Serum (mEq/L)	103	97–108
Calcium, Serum (mg/dL)	9.5	8.7–10.2

Creatinine test: This test measures the creatinine levels in blood or in urine. Creatinine is a waste product made by our muscles as a part of regular, everyday activity.

Normally, our kidneys filter creatinine from our blood and send it out of the body in our urine. This is an important test that is done in diabetes mellitus.

- ❖ For creatinine blood test: A health care professional will take a blood sample from a vein of our arm, using a small needle. After that the needle is inserted, a small amount of blood will be collected into a test tube or vial. We may feel a little sting when the needle goes in or out. This actually takes less than 5 minutes.
- ❖ For creatinine urine test: The health care provider will ask us to collect all urine during a 24 hour period. Our health care provider or a laboratory professional will give us a container to collect and store our samples. A 24 hour urine sample test generally includes the following steps:
- ✓ The bladder is emptied in the morning and the urine is flushed away. The time is recorded.
- ✓ For the next 24 hours the urine is preserved in a container provided.
- ✓ The urine container is stored in a refrigerator or a cooler place.
- ✓ The sample container is then returned to a health provider or the laboratory as instructed.

Sodium test: A sodium blood test measures the amount of sodium in blood. Sodium is a type of electrolyte . Electrolytes are electrically charged minerals that help to maintain fluid levels and the balance of chemicals in our body called acids and bases. It also helps our nerve and muscle work properly.

A sodium blood test may be a part of a test called an electrolyte panel. An electrolyte panel is a blood test that measures sodium, along with other electrolytes, including potassium, chloride, and bicarbonate.

The test can be conducted as

- ❖ A health care professional will take a blood sample from a vein in our arm, using a small needle.
- ❖ After the needle is inserted, a small amount of blood will be collected into a test tube or vial.
- ❖ We may feel a little sting when the needle goes in or out.
- ❖ This usually takes less than five minutes.

Potassium test: A potassium test is used to measure the amount of potassium in our blood. Potassium is an electrolyte that's essential for proper muscle and nerve function. Even minor increases or decreases in the amount of potassium in our blood can result in serious health problems. A potassium test is often performed as a part of a basic metabolic panel, which is a group of chemical tests run on your blood serum.

Prior to the test our doctor may want us to stop taking any medications that could affect the test results. We should ask our doctor for specific instructions prior to our test day. It is performed in a same way like other routine blood tests.

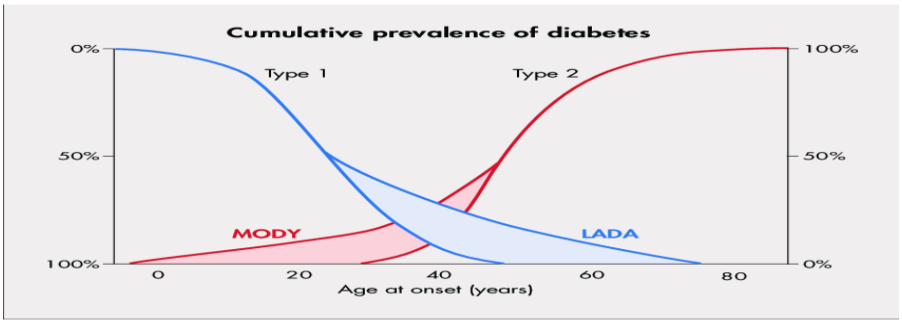
- A site on our arm, usually the inside of our elbow or the back of our hand, will be cleaned with antiseptic.
- A needle will be inserted into our vein. We may feel a sting or the prick of the needle. Blood will be collected into a tube. The band and the needle will be removed and the site will be covered with a small bandage.
- The test generally takes only a few minutes.

Oral Glucose Tolerance Test (OGTT): A glucose tolerance test measures how well your body's cells are able to absorb glucose (sugar) after you consume a specific amount of sugar.

For oral glucose tolerance test:

- A two hour, 75 gram oral glucose tolerance test is used to test for diabetes.
- A health care provider will take a fasting lab draw of blood to test our fasting glucose level first.
- They will then ask us to drink 8 ounces of a syrupy glucose solution that contains 75 grams of sugar.
- We have to then wait in the office for two hours. The health care provider will draw blood at the one and two hour marks.

CONCLUSION



Researchers have been effectively working to investigate the genetic determinants and pathophysiology of MODY by means of various powerful approaches (based on genome and cell biology). These studies have been very promising and insightful which has greatly extended our understanding of both normal and pathologic β-cell biology as well as regulation of insulin secretion in humans. At clinical level, MODY requires correct and accurate methods of diagnosis as well as to distinguish it from diabetes type 1 or type 2 to avoid the unnecessary insulin or sulfonylureas treatment, which may severely affect the patient's health.

In future, translational biology and integrative genomic research and studies on both monogenic and polygenic forms of diabetes are required, which will not only provide novel insights but, also broaden our understanding in terms of pathophysiology, treatment and cure of diabetes. Currently, application of NGS has emerged a key factor to decipher the large amount of known genetic defects underlying pancreatic islet β-cell dysfunction in diabetic patients. As there is increased rate of MODY across the globe, NGS can serve as one of the potential and rapid molecular based diagnostic tool as well as to identify novel genetic etiologies of familial or atypical early-onset diabetes. Transcriptomic and secretomic analysis related to MODY may provide a significant molecular insight into the disease and its procurement. Besides genomic approaches, stem cell technology can also be another novel and effective disease model to study and therapy for the diabetes. Use of pluripotent stem cells (PSCs) can help us to better understand the molecular mechanisms underlying different forms of diabetes (MODY). In addition, metabolomic profiling can also be one of the important approaches to study and differentiate MODY with other diabetes

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The prevalence of LADA is clinically underreported among D.M2 patients and age of onset of diabetes should no longer be considered as a valid way to differentiate diabetes.

2. Patients with LADA who are characterized by autoimmunity to pancreatic beta cells show a clinical phenotypic with anthropometric features that are similar to type 1diabetic patients & differed from those clinically observed in patients with type 2 DM.

3. LADA increases with increasing age decade.

4. D.M.2 patients were more obese & overweight than LADA patients due to insulin resistant in those patient.

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